

# Alexandre Blanc

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## EDUCATION

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- 2021 - 2025 PhD (Physics) at **Paris Sciences et Lettres and Pasteur Institute, Paris, France**  
Behavior modelling using deep autoencoders, biomechanical modelling with the finite element method, neural circuit modelling with rate models.  
Supervised by Jean-Baptiste Masson and Christian Vestergaard.
- 2019 Engineer Degree at **ENSTA, Palaiseau, France**, major in Optimization for Data Science
- 2016 - 2019 Engineer Degree at **Ecole polytechnique, Palaiseau, France**, major in mathematics
- 2014 - 2016 Preparatory Classes at **Lycée Louis-le-Grand, Paris, France**, undergraduate physics, mathematics and computer science

## PROFESSIONAL EDUCATION

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- 2024 Spring School on Optical Imaging and Electrophysiological Recording in Neuroscience, Paris, France
- 2024 PSL-QLife Quantitative Biology Winter School on Learning and Plasticity in Neural Networks, Paris, France
- 2023 PSL Graduate Program Advanced Course in Computational Neuroscience, Paris, France

## PUBLICATIONS

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- Blanc, Alexandre (Dec. 2025). "The Maggot's Behavior, Brain and Body: Computational Modelling in *Drosophila Melanogaster* Larvae". PhD thesis. Paris: Paris Sciences et Lettres.
- Blanc, Alexandre et al. (Apr. 2025). "Statistical Signature of Subtle Behavioral Changes in Large-Scale Assays". In: *PLOS Computational Biology* 21.4, e1012990. ISSN: 1553-7358. DOI: [10.1371/journal.pcbi.1012990](https://doi.org/10.1371/journal.pcbi.1012990).
- de Treder, Eloïse et al. (Sept. 2025). "Feeding State-Dependent Neuropeptidergic Modulation of Reciprocally Interconnected Inhibitory Neurons Biases Sensorimotor Decisions in *Drosophila*". In: *Nature Communications* 16.1, p. 8198. ISSN: 2041-1723. DOI: [10.1038/s41467-025-61805-y](https://doi.org/10.1038/s41467-025-61805-y).
- Laurent, François et al. (July 2024). "LarvaTagger: Manual and Automatic Tagging of *Drosophila* Larval Behaviour". In: *Bioinformatics* 40.7, btae441. ISSN: 1367-4811. DOI: [10.1093/bioinformatics/btae441](https://doi.org/10.1093/bioinformatics/btae441).

## SHORT TALKS AND POSTER PRESENTATIONS

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I gave short talks for:

- Pasteur Institute's QBio group (Dec. 2021)
- the lab meeting at Tomoko Ohyama's lab in Montreal (Dec. 2022)
- Pasteur Institute's Annual Brain Axis Meeting (Mar. 2023)
- the student seminar of Pasteur Institute's Computational Biology Department (Apr. 2023)
- the 2024 March Meeting of the American Physical Association in Minneapolis (Mar. 2024)
- Pasteur Institute's "Brain in the Body" symposium (Nov. 2024)
- the INRIA Workshop on Digital Twins (Nov. 2024)
- the student seminar of INRIA Paris (Apr. 2025)

I presented posters at :

- the 3IA Doctoral Workshop in ANITI, Toulouse (Nov. 2021)
- the annual meeting of the American Biophysical Society in San Francisco (Feb. 2022)
- the 2024 March Meeting of the American Physical Association in Minneapolis (Mar. 2024)
- the Paris Île-de-France Neural Theory and Systems Symposium (Dec. 2024)
- the CNRS' GDR NeuralNet annual meeting (Nov. 2024)

## WORK EXPERIENCE

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### Institut Pasteur, Paris, France

April 2020 - September 2020

Research internship in the Decision and Bayesian Computation Lab supervised by Jean-Baptiste Masson. Deep learning based models of behavioral data from *D. Melanogaster* optogenetic assays.

### Wandercraft, Paris, France

April 2019 - July 2019

Research internship in the Command and Control team. Feedforward control of a lower body exoskeleton for rehabilitation.

## SOFTWARE

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### MaggotUBA



Pytorch-based research software for modelling large scale behavioural assays of *Drosophila* larvae in an unsupervised fashion.

### Pytrxmat



Pure python package to handle manipulation of behaviour data files.

## OTHERS

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|------------------|-----------------------------------------------------------------------------------------------------|
| Languages        | French (native), English (fluent), Spanish (basic)                                                  |
| Campus life      | Graduate students' elected representative for the neuroscience department of the Pasteur Institute. |
| Volunteering     | I organize summer camps for teenagers with the Scouts et Guides de France.                          |
| Hobbies & Sports | Many ; I keep returning to running.                                                                 |

Last updated: August 5, 2026